Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/03/2012
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Radiation and Gliomagenesis: A Sensitive Model System to Evaluate the Tumorigenic Potential of HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	02/01/2010	**End Date:**	12/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2013 per NSSC info (Ed., 1/24/2013) NOTE: to be extended past 1/31/2013, per A. Chu/ARC (Ed., 12/4/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bachoo, Robert (University of Texas Southwestern Medical Center) Boothman, David (University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX10AE08G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There is a paucity of systematic studies carried out with well-defined model systems to evaluate the tumorigenic potential of HZE particles that are presumed to pose a significant cancer risk to astronauts. We have shown that these particles induce DNA double-strand breaks (DSBs) that are repaired slowly and incompletely and trigger persistent DNA damage signaling events. In order to evaluate the carcinogenic consequences of such unrepaired DNA lesions, we propose to utilize genetically engineered astrocytes/neural stem cells (NSCs) as well as transgenic mouse models representing progressive steps in the development of Glioblastoma Multiforme (GBM). These lethal brain tumors are the third leading cause of cancer-related death among adults aged 30 to 50 years that are known to be induced by ionizing radiation. The past year has seen unprecedented advances in the genomic analyses of adult GBM tumors by the Cancer Genome Atlas Network, which reveal that these tumors have radically altered genomes with 5 key genetic changes dominating: loss of Ink4a, Arf, p53, or PTEN and amplification of EGFR (especially the constitutively active EGFRvIII). We have developed astrocyte and NSC lines as well as mouse models with targeted deletions of these genes in logical combinations representing the progression of primary or secondary GBMs. This genetic progression model, though reductionist in approach, provides a realistic framework for evaluating and understanding how known genetic mutations may cooperate with charged particle radiation and contribute to the initiation and progression of high-grade gliomas. We hypothesize that poorly repaired DNA damage produced by charged particles will result in malignant progression towards brain cancer, and that this progression can be accelerated by predisposing oncogenic activations/tumor suppressor losses. More importantly, these "pre-initiated" murine astrocytes/NSCs will allow us to rapidly quantify the transforming potential of charged particles relative to gamma rays both in vitro and in vivo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This grant seeks to understand cancer risks (especially the risk of brain cancers) from particle radiation in outer space. However, charged particles are also being used in the clinic for targeted tumor therapy. In addition to evaluating cancer risks to astronauts, our project will also contribute significantly to our understanding of cancer cell killing by charged particles in the clinic and can shed light on the risk of secondary cancers from particle therapy.
Task Progress:	Brief summary of progress--In experiments carried out during the third year of the project, we have been able to obtain quantitative estimates of the RBE for transformation by Fe ions relative to X-rays. Estimates using the Ink4a;4b;Arf cohort of mice indicate an RBE of at least 4. We are awaiting results from the irradiated p53;PTEN cohort in order to confirm these RBE estimates. In addition we find that amplification of the proto-oncogene MET is a key event in Fe-induced gliomagenesis. The exact role(s) of MET in this process is being elucidated. Detailed summary of progress. Our currently-funded NASA project essentially has two components: 1) An 'in vitro' component where primary murine astrocytes and neural stem cells (NSCs) with glioma relevant genetic alterations are assayed for transformation after particle radiation. Briefly, we find that Fe ions are more tumorigenic than gamma rays and that loss of the Ink4a/4b/Arf tumor suppressor locus is a critical event in HZE-induced transformation. The in vitro work is published and results from this project have guided the development of the second component of our project. 2) For the 'in vivo' component, we have used genetically engineered mice in which deletions of the most critical, glioma-relevant tumor suppressor genes (Ink4a, Ink4b, Arf, p53 and/or PTEN) can be targeted to the mouse brain. The Ink/Arf and p53 deletions are generally mutually exclusive in GBM, hence we are using two distinct cohorts of mice. These models, in a reductionist manner, represent progressive steps in gliomagenesis, and based on our published in vitro work, we hypothesized that they may have heightened sensitivity to transformation by particle radiation. We compared brain tumor development in these models after irradiation with 1Gy or 0.5Gy of 600 MeV/n Fe ions, in comparison with 1Gy or 4 Gy of X-rays. Brain tumor incidence observed at 8 months with loss of Ink4a and Arf alone was low after exposure to X-rays (4Gy) or Fe ions (1Gy). Importantly, the incidence of brain tumors increased significantly with additional loss of Ink4b, corroborating our in vitro studies which showed that Ink4b is an important barrier to radiation-induced transformation. The tumor incidence was the same (25%) with 1Gy of Fe ions and a 4-fold higher dose of X-rays, suggesting a RBE of 4 for transformation by Fe ions. Finally, genomic analyses indicate that MET amplification is a critical event in HZE-induced gliomagenesis and that MET signaling supports a GBM cancer stem cell phenotype.
Bibliography Type:	Description: (Last Updated: 12/09/2025)
Articles in Peer-reviewed Journals	Mukherjee B, Tomimatsu N, Amancherla K, Camacho CV, Pichamoorthy N, Burma S. "The dual PI3K/mTOR inhibitor NVP-BEZ235 is a potent inhibitor of ATM- and DNA-PKCs-mediated DNA damage responses." Neoplasia. 2012 Jan;14(1):34-43. PubMed PMID: 22355272 , Jan-2012
Articles in Peer-reviewed Journals	Tomimatsu N, Mukherjee B, Deland K, Kurimasa A, Bolderson E, Khanna KK, Burma S. "Exo1 plays a major role in DNA end resection in humans and influences double-strand break repair and damage signaling decisions." DNA Repair (Amst). 2012 Apr 1;11(4):441-8. Epub 2012 Feb 11. PubMed PMID: 22326273 , Apr-2012
Articles in Peer-reviewed Journals	Morris C, Tomimatsu N, Richard DJ, Cluet D, Burma S, Khanna KK, Jalinot P. "INT6/EIF3E interacts with ATM and is required for proper execution of the DNA damage response in human cells." Cancer Research. 2012 Apr 15;72(8):2006-16. PubMed PMID: 22508697 , Apr-2012
Articles in Peer-reviewed Journals	Costelloe T, Louge R, Tomimatsu N, Mukherjee B, Martini E, Khadaroo B, Dubois K, Wiegant WW, Thierry A, Burma S, van Attikum H, Llorente B. "The yeast Fun30 and human SMARCAD1 chromatin remodellers promote DNA end resection." Nature. 2012 Sep 27;489(7417):581-4. Epub 2012 Sep 9. http://dx.doi.org/10.1038/nature11353 ; PubMed PMID: 22960744 , Sep-2012
Articles in Peer-reviewed Journals	Raghavan P, Tumati V, Yu L, Chan N, Tomimatsu N, Burma S, Bristow RG, Saha D. "AZD5438, an inhibitor of Cdk1, 2, and 9, enhances the radiosensitivity of non-small cell lung carcinoma cells." Int J Radiat Oncol Biol Phys. 2012 Nov 15;84(4):e507-14. Epub 2012 Jul 12. http://dx.doi.org/10.1016/j.ijrobp.2012.05.035 ; PubMed PMID: 22795803 , Nov-2012
Articles in Peer-reviewed Journals	Kobayashi J, Fujimoto H, Sato J, Hayashi I, Burma S, Matsuura S, Chen DJ, Komatsu K. "Nucleolin participates in DNA double-strand break-induced damage response through MDC1-dependent pathway." PLoS One. 2012;7(11):e49245. Epub 2012 Nov 7. http://dx.doi.org/10.1371/journal.pone.0049245 ; PubMed PMID: 23145133 , Nov-2012